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Tabula Magna

User Manual

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1 Getting started

1.1 Overview

Tabula Magna is a MIDI stop-tab unit for virtual pipe organs. It is an add-on to your organ console: a flat box with an adjustable stand. It presents **40 physical stop tabs** and **5 additional display widgets**, and connects to your computer over a single USB cable, where it is used as a MIDI device by **Hauptwerk** (or any other MIDI and labeling-capable organ software).
A colour display sits behind the tab rail. It shows the stop names, colours and states sent by Hauptwerk over MIDI, so the tab labels are not fixed: whatever *sample set* you load, the labels and colours follow automatically. The same display can also show the current sample-set name and other information in the widget strip along the bottom.
Tabula Magna needs no driver and no configuration program installed on your computer. It is configured from Hauptwerk itself (Section 2); the unit's own settings, reached over its built-in Wi-Fi, are there for advanced adjustments and as a handy reference while you set up Hauptwerk (Section 3).

1.2 Features

- **40 stop tabs** arranged in 5 rows of 8, each lit and labelled from MIDI (Hauptwerk LCD32 displays).
- **5 panel widgets** — a wide strip along the bottom for sample-set names, crescendo, swell and similar read-outs.
- **Hall-sensor keying** — each tab is read contactlessly, with no moving electrical contacts to wear out.
- **Colour LCD labelling** — stop names and background colours follow the loaded Hauptwerk sample set.
- **USB MIDI** — class-compliant, plug-and-play on Windows, macOS and Linux. The device appears by its own name, unique per unit.
- **Two MIDI binding styles per tab** — *Note On/Off* or *NRPN* — configurable independently for what the tab listens to and what it sends.
- **MIDI Learn** — bind any incoming message to a tab by simply sending it, without typing numbers.
- **Per-tab behaviour** — *ON/OFF*, *Momentary* or *Disabled*, per tab or for all tabs at once.
- **Smart drag** — sweep a finger across a row to set several stops in one gesture.
- **Built-in Wi-Fi settings page** — a live reference and test bench from a phone or laptop, plus the unit's advanced adjustments, with no extra software.
- **Sample-set-driven settings** — a Hauptwerk sample set can reconfigure the unit's fonts, colours and behaviour when it loads (Section 4.3).

1.3 What you received

Your unit is delivered **already registered**, so it works out of the box. The details on the registration card are only needed again if the software is reinstalled — for example by writing a fresh system image to a new SD card.

- 1× Tabula Magna unit.
- 1× USB cable (USB-A to USB-A).
- 1× external power supply.
- 1× registration card, carrying this unit's **Device ID**, its **unlock code** and the password for the unit's Wi-Fi access point (Section 3).

Note: Keep the registration card. Its **unlock code** is unique to your unit and is needed if the software is reinstalled.

1.4 Specifications

- Dimensions: 415×275×80 mm.
- Weight: 3.2 kg.
- Power supply: 5 V DC, 5 A.

1.5 Installation

1.5.1 Positioning

Place the unit to the side of the manuals, where you want it on your organ (or on a desk) — typically on the left or the right of them. If you have more than one unit, place them to either side. Its stand is adjustable, so you can set the viewing angle. It is standalone: it needs only power and one USB connection to the computer running Hauptwerk. The USB connection carries MIDI data only. The unit always runs from its external power supply and cannot be powered over USB, so connect the supplied power supply whenever you use the unit.

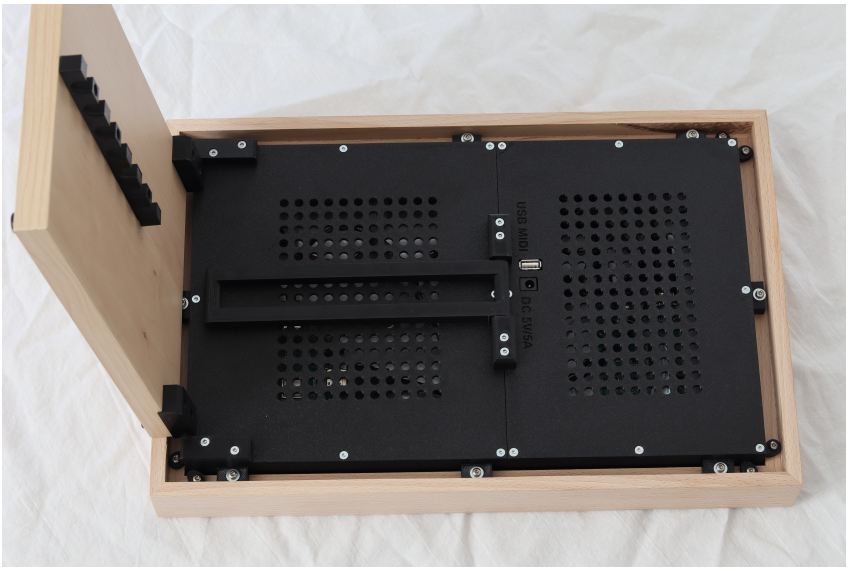


Figure 1: Rear/side view of the unit showing the power inlet and the USB port.

1.5.2 Connecting

1. Connect the unit to your computer with the supplied USB cable (USB-A to USB-A).
2. Connect the supplied power supply to the unit and plug it into a mains outlet.

3. The unit starts up. This takes about 30 seconds.

> ⚠ **Important:** The unit always requires its external power supply. The USB cable carries MIDI data only and cannot power the unit.

On first connection, your computer will recognise a new MIDI device. On Windows it is listed as:

`Tabula` *`XXXXXX`*

where `XXXXXX` is this unit's Device ID, so several units on the same system remain distinguishable.

> Once a unit has been used on a given computer, the computer remembers it. If the unit is later plugged into a different USB port or computer, it should be recognised again as the same device.

2 Setting up in Hauptwerk

Hauptwerk is where the unit is configured. The unit ships with defaults that suit almost every use case, so you normally never change anything on the unit itself. All the work is done once, in Hauptwerk, by following the four steps below.

> ℹ **Note:** Everything you set in Hauptwerk is stored **per sample set** — Hauptwerk's dialogs say "for this organ only". Each sample set keeps its own stop labels, colours and MIDI bindings, so loading a different organ relabels the tabs and reconfigures the unit automatically. This is the point of the unit.

> ℹ **Note:** Leave the unit's own settings at their defaults. They are chosen so that the four steps below work with no adjustment. Change them only if you have a specific reason to (Section 3). The settings app's home page (Section 3.2) is a convenient reference for the addresses you need in Step 2.

2.1 Step 1 — MIDI ports

You tell Hauptwerk to use Tabula Magna as both an output (to light the tabs) and an input (to read the tabs you press).

> ⚠ **Important:** Switch the unit on and wait for it to finish starting (about 30 seconds) *before* Hauptwerk loads its sample set. If the unit is still starting when the sample set loads, Hauptwerk may not find it, and the stop labels and states will not reach the tabs.

1. In Hauptwerk, open **General settings → MIDI ports** (availability depends on your Hauptwerk version).
2. Enable the *Tabula* MIDI input port, so Hauptwerk receives your tab presses.
3. Enable the *Tabula* MIDI output port, so Hauptwerk can send stop labels, colours and states to the unit.

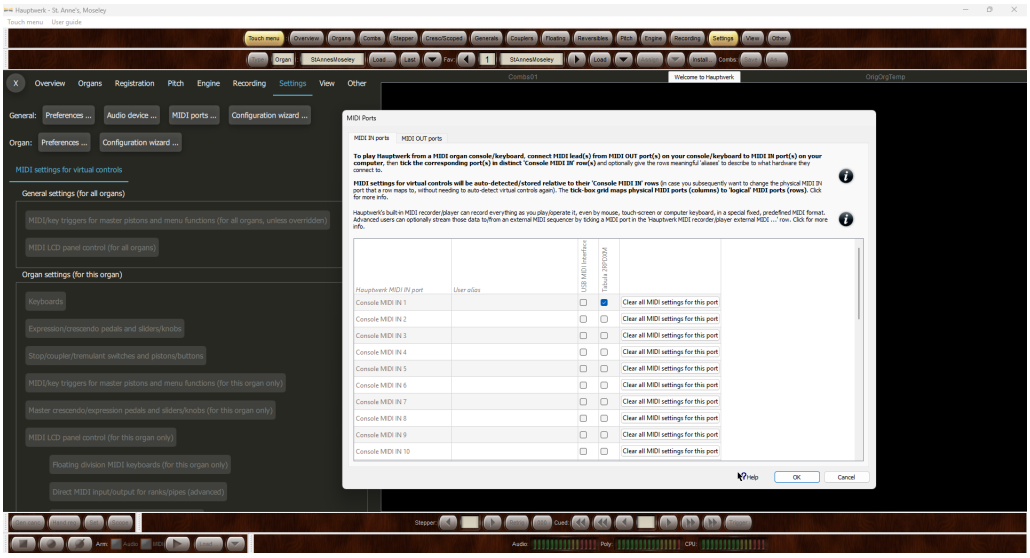


Figure 2: Hauptwerk's MIDI ports dialog with the Tabula ports enabled.

2.2 Step 2 — Stop labels

Hauptwerk cannot label tabs directly: it pushes short text to numbered **LCD32 displays**. In Hauptwerk, open **MIDI LCD panel control (for this organ only)**. This is a flat list of LCD instructions, each with an *address* and two lines of text. You can add and remove instructions, but you cannot reorder the list — so keep it tidy by creating the instructions in address order: address 1 for the first tab, address 2 for the second, and so on. The default layout uses addresses **1–40** for the tabs and **41–45** for the panel widgets (Section 4.1).
Address **128** is reserved for the unit's own settings commands (Section 4.3). If you use them, add those instructions *last*, so the label list stays in order.

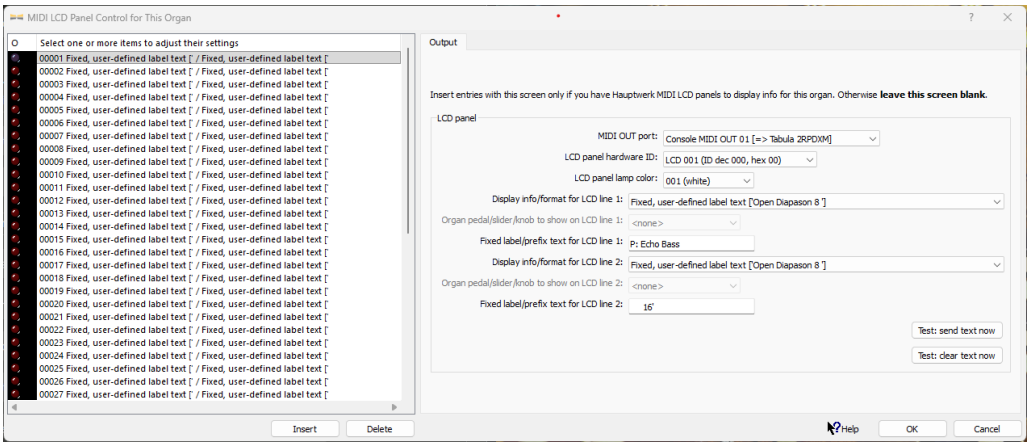


Figure 3: One LCD instruction in Hauptwerk's MIDI LCD panel control list, showing the display address and the two lines of stop text.

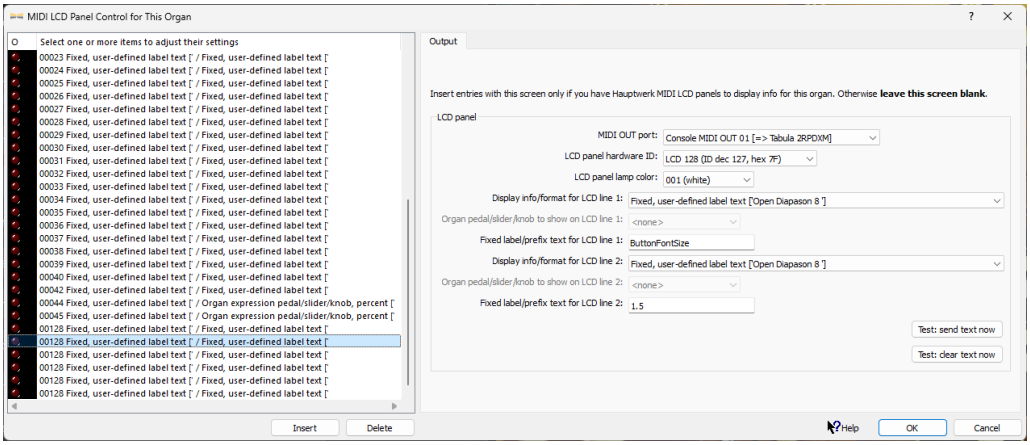


Figure 4: An LCD instruction for address 128, carrying a settings command.

Note: The addresses here are the same numbers the unit uses. There is no "off-by-one" to remember — set 12 in Hauptwerk and the unit's tab 12 follows.

2.3 Step 3 — Pairing the tabs

Once the labels are correct, pair each tab with its stop. Use Hauptwerk's *listen* (auto-detect) function: Hauptwerk captures the messages the unit sends when you press the tab.

1. Right-click the stop in Hauptwerk and choose its auto-detect (*listen*) option, so Hauptwerk waits for incoming MIDI.
2. Press the matching tab on the unit **twice** — once to switch it on and once to switch it off. Hauptwerk expects to see both the ON and the OFF message.

The unit sends **NRPN on MIDI channel 2** by default, so this is what Hauptwerk captures. This fills in the stop's *input* — what Hauptwerk listens to.

Important: Always press the tab twice, so Hauptwerk receives both the ON and the OFF message. If it sees only one, it treats the stop as a momentary button and behaves incorrectly.

2.4 Step 4 — Outgoing MIDI

The reverse direction — what Hauptwerk sends to light the tab — cannot be learned and must be set by hand. It is quick, because the number is already known from Step 3.

1. Right-click the stop and choose **Adjust MIDI/trigger settings manually for stop: *xxx***.
2. Under **Primary input**, the NRPN number captured in Step 3 is already filled in. Note that number.
3. Under **Primary output**, choose the *Tabula* output card, set the event type to **NRPN**, MIDI **channel 1**, and enter the *same* NRPN number.

Note: The unit listens on channel 1 and sends on channel 2, so in Hauptwerk the stop's *input* is channel 2 and its *output* is channel 1 — with the *same* NRPN number on both.

Repeat Steps 3 and 4 for every stop you want to use.

3 The unit's own settings (Wi-Fi)

The unit also has its own settings, reached over its built-in Wi-Fi. The defaults are correct for almost every organ, and everything in Section 2 is done in Hauptwerk, not here. **You should not need to change any MIDI setting in the app**, and normally you will not. But the app is far from useless in common use: it is a useful companion while you set up Hauptwerk, and it is the home of the few adjustments that only exist on the unit. Section 3.2 lists the common reasons to open it.
These settings are saved on the unit and **persist across reboots**. The app is not the main workflow: prefer configuring MIDI from Hauptwerk (Section 4.3) whenever you can.

3.1 Joining the unit's Wi-Fi

1. Power on the unit and wait about 30 seconds for it to finish starting.
2. On your phone or computer, open the Wi-Fi settings and look for a network named:

   `Tabula-`*`XXXXXX`*

   where `XXXXXX` is this unit's Device ID.
3. Join it, using the password printed on the registration card.
4. Open a web browser and go to:

   `192.168.4.1`
5. The unit's settings page opens.

Note: While you are connected to the unit's Wi-Fi, your phone or computer will not have internet access. That is normal — disconnect from the unit's network afterwards to return to your usual connection.

Important: The Wi-Fi access point is always on. It is the unit's only configuration channel and remains available at all times.

3.2 The settings app

The settings application runs at `192.168.4.1` (the root of the unit's web server). It holds the unit's own settings. It is split into the following pages, reachable from the navigation bar at the top.

3.2.1 Common uses

You do not need the app to *change* anything, and you should leave the MIDI settings alone. Even so, it earns its place in normal use. There are three common reasons to open it:

1. **A cheat sheet while you set up Hauptwerk.** The home page shows the whole unit at a glance: every tab and panel widget laid out in the same arrangement as on the instrument, each with its **address**. Keep it open on a phone beside you as you enter addresses in Hauptwerk (Section 2) instead of counting tabs or guessing.
2. **Trying a look before committing to it in Hauptwerk.** Font family, font size and label formatting can be changed here and seen on the unit immediately. Find the combination you like first, then set the same values in the Hauptwerk command that makes them permanent (Section 4.3).

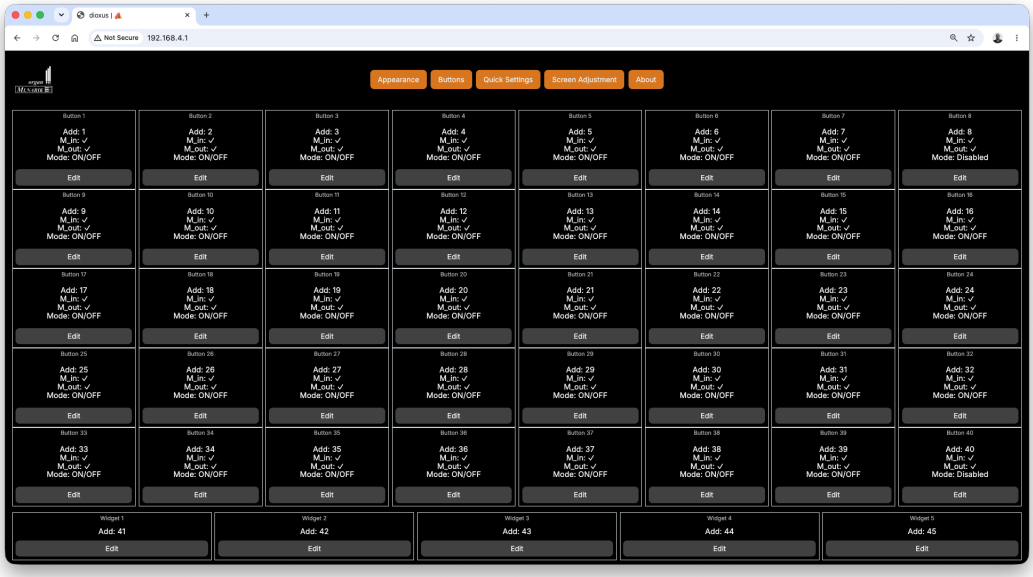


Figure 5: The settings home page, showing the grid of tab tiles and the panel widget tiles with their Edit buttons.

Home (/) A tile for every tab (1–40) and every panel widget (1–5), each showing its ID, LCD address and an **Edit** button.

Appearance Fonts, label sizes, how LCD32 text is formatted, and whether the grid lines are drawn. (Section 3.5.)

Buttons Physical-tab input mode and the calibration tool. (Section 3.6.)

Quick Settings Bulk tools: apply one address range, one MIDI binding or one behaviour to every tab at once. (Section 3.7.)

Screen Adjustment Fine nudges to line the display up with the physical tabs. (Section 3.9.)

About Versions and device identity — the diagnostics surface. (Section 3.10.)

> **Note:** Changes are saved automatically. Each page's **Apply** button commits that one setting; there is no separate global save. The **Default** button on a row restores that row's factory value.

3.3 Editing a tab

Press **Edit** on a tab tile on the home page to open the **Edit Button** page, with four sections.

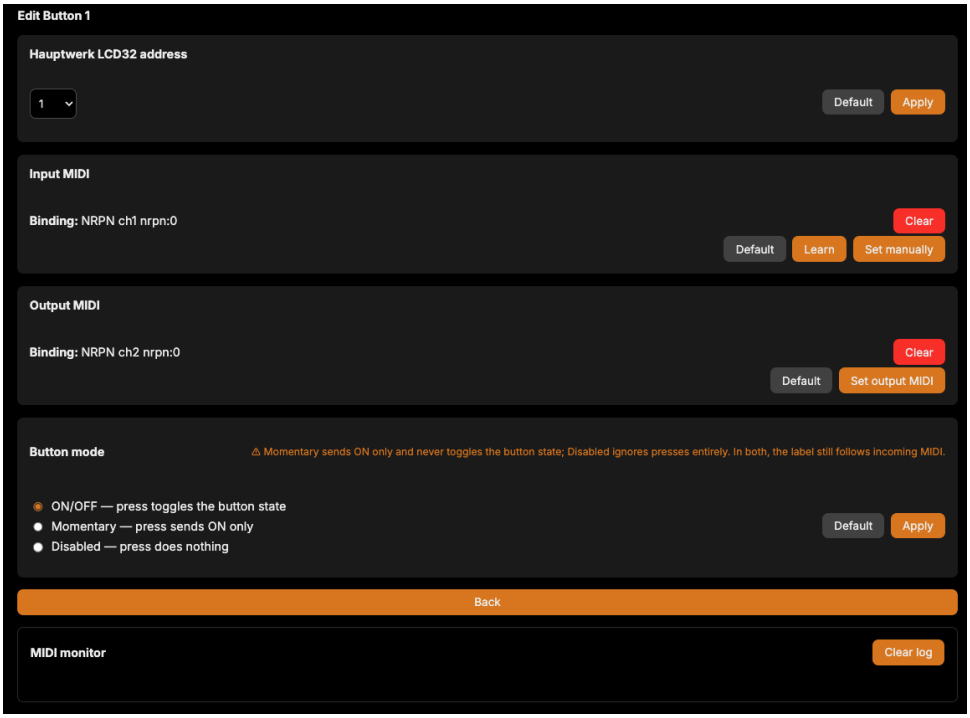


Figure 6: The Edit Button page: LCD32 address, Input MIDI, Output MIDI, and Button mode.

3.3.1 Hauptwerk LCD32 address

The address this tab listens to for its label and colour. Range **1–127**. **Default** restores the natural address for the tab (tab 1 → 1, tab 2 → 2, and so on).

3.3.2 Input MIDI

The MIDI message that *actuates* this tab (what Hauptwerk sends when it wants to switch the stop on or off). The current binding is shown as text; press **Clear** to remove it.

- **Learn** — bind automatically by playing the message from Hauptwerk.
- **Set manually** — choose the type, channel and number by hand.

3.3.3 Output MIDI

The MIDI message this tab *sends* when you press it. Configured exactly like the input side. (**Learn** does not apply to the output — use **Set output MIDI**.)

3.3.4 Button mode

How a press on this tab is interpreted — see Section 3.8. **Default** restores *ON/OFF*.

3.3.5 MIDI types

Both input and output bindings choose one of two *types*:

Note On/Off A MIDI channel (1–16) and a note number (0–127). Switching on sends a Note On; switching off sends a Note Off. The velocity is fixed by the unit.

NRPN A MIDI channel (1–16) and a 14-bit NRPN number (0–16383). Switching on sends the full NRPN *activate* sequence; switching off sends the *deactivate* sequence. The unit builds the underlying control-change messages for you.

> You only ever configure *one* binding per direction. There is no separate "on" and "off" message to set: the unit derives both from the single binding.

3.3.6 The MIDI monitor

Below the binding sections on the Edit Button page is a **MIDI monitor**: a live list of the most recent MIDI messages the unit has received, in human-readable form. The newest line is highlighted. **Clear log** empties it. It is a diagnostic tool — useful to confirm that Hauptwerk is actually sending what you expect.

3.4 Editing a panel widget

Each panel widget works the same way as a tab, but with fewer options. Press **Edit** on a widget tile to set its **LCD32 address** (range **1–127**; **Default** restores 41–45). Panel widgets have no MIDI binding and no mode — they only display what Hauptwerk sends them, plus the MIDI monitor for diagnostics.

3.5 Appearance

The **Appearance** page controls how labels look on the unit.

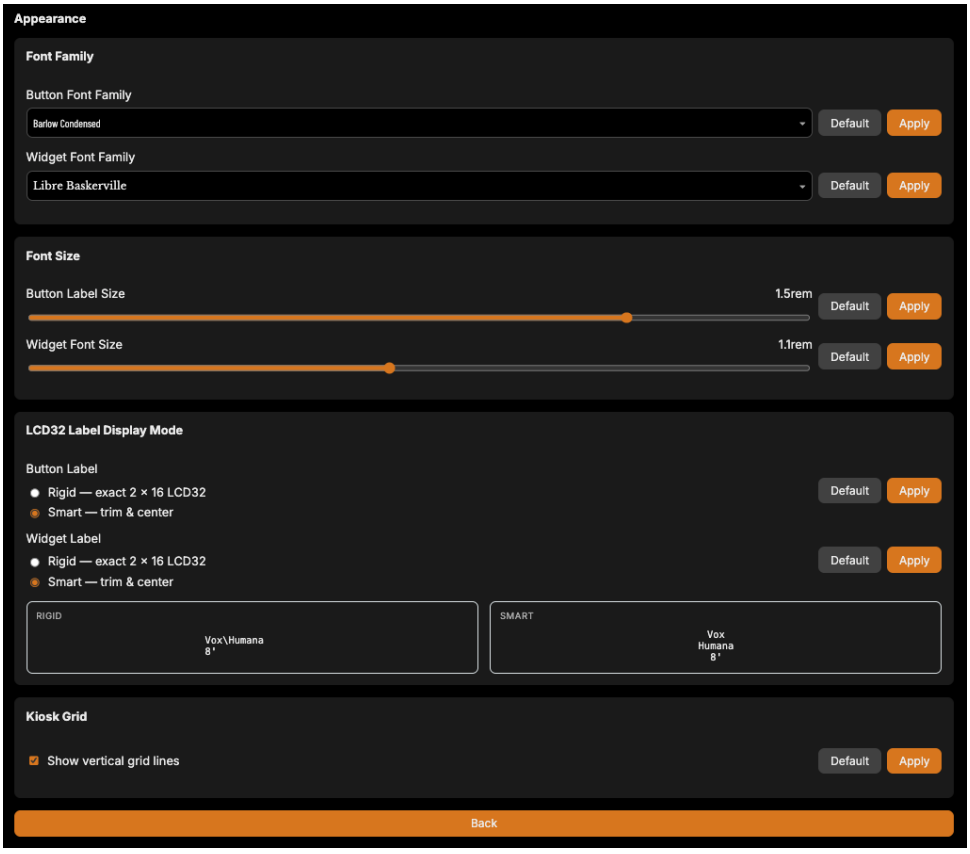


Figure 7: The Appearance page: font family, font size, label display mode and the kiosk grid.

3.5.1 Font family

Choose the typeface used for **tab labels** and for **panel widget labels**, independently. Choose from the built-in list; **Default** returns to *Noto Sans*.

3.5.2 Font size

A slider adjusts the **tab label** size and the **panel widget** size, independently, from **0.5** to **1.8** (relative units). **Default** returns to the standard size.

3.5.3 LCD32 label display mode

Hauptwerk sends each display a raw two-line block of 16 characters per line. This setting decides how that raw text is turned into the label you see. It is set independently for tabs and for panel widgets:

Smart *(default)* Trims spaces from each line, centres the text, and treats a backslash \ as a forced line break. Empty lines are removed.

Rigid Reproduces the raw two-line block exactly, preserving every space, with no centring or trimming.

> **Note:** In Smart mode, a genuine backslash inside a stop name is indistinguishable from a forced line break. Use Rigid mode, or avoid a literal backslash in stop names.

3.5.4 Vertical lines

A checkbox turns the vertical separator lines between tabs on the unit display on or off. **Default** shows them.

3.6 Buttons

The **Buttons** page configures the *physical* tabs.

3.6.1 Physical button input mode

How the unit reads a press on the physical tabs:

Smart drag *(default)* Sweeping a finger across a row applies one action to every tab touched. The action is decided by the *first* tab you touched: if it was off, the whole sweep switches the touched tabs on; if it was on, the whole sweep switches them off. Tabs already in the right state send nothing. A single tap still behaves like an ordinary toggle.

Toggle Each tab is independent: pressing it flips its state. A held tab toggles exactly once.

3.6.2 Button calibration

Hall sensors are read against per-tab thresholds. If a tab becomes unreliable (needing a hard press, or triggering when barely touched), recalibrate:

1. Press **Start calibration process**.
2. Press **every** tab for at least one second, one at a time, releasing each one afterwards. A running count shows how many tabs have been detected.
3. Press **Done**. The thresholds are stored to the unit.

Tabs you do not press keep their existing thresholds, so you may calibrate only the tabs that need it.

> **Note:** Dismissing the calibration dialog (pressing outside it) *cancels* the run and discards the changes. Press **Done** to save.

> **Note:** **Default** on the calibration row resets all thresholds to the factory values.

3.7 Quick Settings

Quick Settings applies one configuration to *all* tabs at once. Each section has its own **Apply** button, and each carries a warning that it will overwrite the per-tab settings.

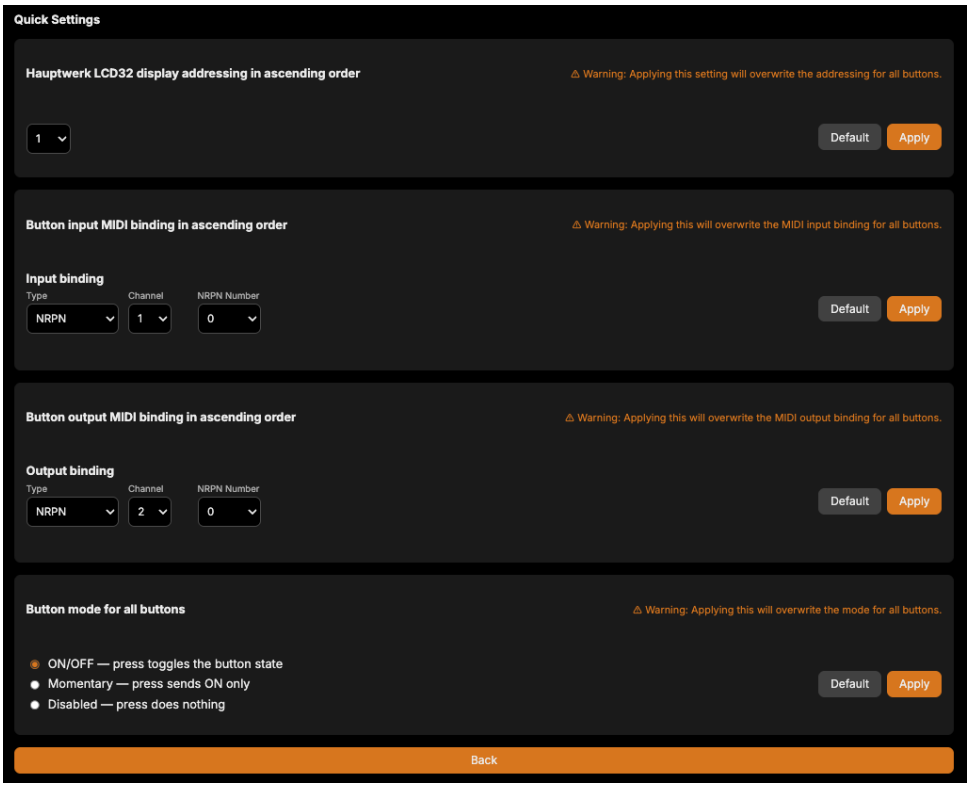


Figure 8: The Quick Settings page, with the address, MIDI binding and mode sections.

Hauptwerk LCD32 display addressing Sets a contiguous block of addresses in ascending order, starting from the value you choose (1–84, so all 45 addresses fit within 1–128). Tab 1 gets the base address, tab 2 the next, and so on; the panel widgets continue after the last tab.

Button input MIDI binding in ascending order Applies one binding to every tab, incrementing the number per tab: tab 1 gets the chosen number, tab 2 the next, and so on. Configured with the same editor as a single tab (Section 3.3).

Button output MIDI binding in ascending order The same, for what the tabs send.

Button mode for all buttons Sets every tab to the same behaviour — see Section 3.8.

3.8 Button behaviour modes

Independent of the input mode above, each tab has a **mode** that decides what a press actually does. It is set per tab (Edit Button page) or for all tabs at once (Quick Settings).

ON/OFF *(default)* A press toggles the tab's state and sends the matching MIDI output. The label follows that state.

Momentary A press sends only the *activate* message and **never** changes the stored state. Release sends nothing. The label therefore keeps following incoming MIDI, which is ideal for groups where Hauptwerk owns the selection (for example a set of *radio-button* stops). Works with both Note On/Off and NRPN output, and for both physical and on-screen tabs.

Disabled The tab is inert. Pressing it (physically or on screen) does nothing, and incoming MIDI bound to it is ignored, so it never lights up. Its label still follows incoming LCD32 text, so a name or colour can be shown without the tab ever toggling.

3.9 Screen Adjustment

Because the display sits behind a physical frame, it may be a fraction of a millimetre out of alignment with the tabs. **Screen Adjustment** nudges the whole visible area to correct this.

Horizontal Offset Shifts the visible area left or right, in sub-millimetre steps. Range −1 to +1 mm. Positive moves right.

Vertical Offset Shifts the visible area up or down. Range −1 to +1 mm. Positive moves down.

The change is applied to the display as soon as you release the slider.

3.10 About

The **About** page is the unit's diagnostics surface — the single place to read what this unit is running, so it can be relayed to your supplier if support is needed.

App The software version installed on the unit.

Hardware The Raspberry Pi board model the unit is running on.

Config The configuration schema version.

License The license scheme version, the **Device ID**, and whether the unit is registered.

Copy diagnostics copies all of this to the clipboard in one block, ready to paste into an e-mail to your supplier.

4 Reference

The earlier sections cover everyday use. This reference gathers the background material: display addressing, two-way MIDI, the commands a sample set can send, the factory defaults, and troubleshooting.

4.1 LCD32 displays and addressing

Hauptwerk cannot label tabs directly: it pushes short text blobs to numbered **LCD32 displays**. Each Tabula Magna tab and panel widget is bound to one such address.

- Hauptwerk lets you choose display addresses **1–128**.
- On the unit, addresses are assignable **1–127**. Address **128** is reserved for the settings channel (Section 4.3) and is never shown as a label.
- The default layout assigns addresses **1–40** to the tabs and **41–45** to the panel widgets.

Element	Count	Default addresses
Stop tabs	40	1–40
Panel widgets	5	41–45
Settings channel	—	128 (reserved)

Note: The addresses shown in this settings app are the same numbers you enter in Hauptwerk. There is no "off-by-one" to remember — set 12 in Hauptwerk and set 12 here.

4.2 Two-way MIDI behaviour

It helps to think of each tab as having two independent halves:

Input What the tab *listens* to. When Hauptwerk sends this message, the tab changes state (and its on-screen highlight updates). Input never triggers the tab's own MIDI output, so there is no feedback loop.

Output What the tab *sends* when you press it. This is what tells Hauptwerk (or any other software) that you operated the stop.

Because they are independent, you can, for example, let Hauptwerk drive the *state* while the tab simply reports presses, or bind both halves to the same message for a straightforward loop.

4.2.1 NRPN notes

For NRPN, the unit understands shortened message sequences as well as the full one: it accepts a sequence missing the final control change, or even a bare value message that reuses the previously selected parameter. This makes it compatible with a wide range of Hauptwerk builds and third-party senders. It only reacts to the NRPN-related control changes, so unrelated MIDI interleaved between them does not disturb it. A value of 64 or more switches the tab on; less than 64 switches it off.

4.2.2 Bulk changes

When many tabs change at once (a sweep, or a burst from Hauptwerk), the unit deliberately spaces its outgoing messages slightly so nothing is dropped on the way to Hauptwerk. This is imperceptible for a bulk action and needs no configuration.

4.3 Driving the unit from Hauptwerk

A Hauptwerk sample set can reconfigure the unit *by itself*: when the sample set loads, it sends its own appearance and behaviour settings to the unit, which reconfigures instantly. This lets each sample set carry its own look (*e.g. a specific font or colour scheme*) without anyone opening the settings app.

4.3.1 How to enable it

1. In Hauptwerk, create an **LCD32 display** at **address 128**. This address is reserved for settings and is never shown as a label.
2. Send a two-line text command to it: **line 1** is the command name, **line 2** is the value.

For example, to set the tab font to *EB Garamond*:

Line 1	Line 2
`ButtonFont`	2

- Commands and values are **case-sensitive** and must be spelled exactly. Each command has only one accepted spelling.
- Leading and trailing spaces are ignored.
- An unknown command or value does nothing.
- Settings take effect immediately and are **not stored**. They last until the unit restarts or the sample set sends them again. Hauptwerk re-sends its displays when a sample set loads, so a sample set re-applies its own settings automatically.

Note: These runtime settings override the web UI while a sample set is loaded, but they never change what the web UI saved. Unload the sample set (or restart) to return to the web-UI settings.

4.3.2 Command reference

Appearance

Line 1	Line 2	Effect
`ButtonFont`	font number	Tab label font
`WidgetFont`	font number	Panel widget label font
`ButtonFontSize`	number (0.5–1.8)	Tab label size
`WidgetFontSize`	number (0.5–1.8)	Panel widget label size
`ButtonLabelMode`	`Rigid` or `Smart`	Tab label formatting
`WidgetLabelMode`	`Rigid` or `Smart`	Widget label formatting
`GridLines`	`On` or `Off`	Show the kiosk grid lines

Buttons

Line 1	Line 2	Effect
`InputMode`	`Toggle` or `Smart`	Physical-tab input mode
`ButtonMode`	`<button> <mode>`	Set one tab's behaviour

For `ButtonMode`, `<button>` is the tab number **1–40** (the number shown in the settings app) and `<mode>` is `OnOff`, `Momentary` or `Disabled`. Set one tab at a time.

Font numbers

No.	Font	No.	Font
1	Barlow Condensed	7	Libre Franklin
2	EB Garamond	8	Noto Sans
3	IBM Plex Sans Condensed	9	Oswald
4	Inter	10	Roboto Condensed
5	`monospace`	11	Source Sans 3
6	Libre Baskerville		

4.3.3 Examples

Line 1	Line 2	Result
`ButtonFont`	`2`	Tabs use EB Garamond
`WidgetFont`	`8`	Widgets use Noto Sans
`ButtonFontSize`	`1.2`	Tab label size becomes 1.2
`ButtonLabelMode`	`Smart`	Tabs use Smart formatting
`GridLines`	`Off`	Hide the grid lines
`InputMode`	`Smart`	Physical tabs use smart drag
`ButtonMode`	`12 Momentary`	Tab 12 becomes momentary

4.3.4 Resetting the unit (Tabula restore)

A sample set can also reset the unit to its *built-in defaults* before it applies its own settings. This is a single, short instruction — the four bytes `F0 7D 02 F7` — which clears every label and colour, returns every tab to its normal ON/OFF behaviour, and restores the default fonts, label modes, grid lines and input mode. Like the settings above, it only affects the running unit and never changes what the web UI saved.

> **Note:** You normally do not need to deal with this instruction: it is sent automatically by Hauptwerk when a sample set is loaded, so every sample set starts from a clean state.

4.4 Using other organ software (raw SysEx)

The unit is not tied to Hauptwerk. Any program that can send MIDI to the unit's output port and receive MIDI from its input port can drive it — for example *GrandOrgue*, *Sweeling*... There are two independent halves to set up:

Real-time control works exactly as described in Section 4.2: bind each tab's **input** (what the tab listens to) and **output** (what it sends when pressed) in the unit's settings app (Section 3.3), choosing *Note On/Off* or *NRPN*. The other program's own auto-detect can capture the unit's output, and the unit's **Learn** can capture the program's output in the other direction.

Labels are the part Hauptwerk normally handles for you. Hauptwerk labels the tabs by pushing **LCD32 SysEx** to them (Section 4.1); other programs do not speak that protocol, so you send the same SysEx frames yourself — as a **raw SysEx file**, or from a small script — before you start playing. The unit keeps the labels until it restarts or receives new ones.

The rest of this section is the concise reference for those frames.

4.4.1 The label frame (39 bytes)

A label is one 39-byte SysEx frame:

Byte(s)	Value	Meaning
1	`F0`	SysEx start
2	`7D`	Manufacturer / device id
3	`01`	"LCD32 label" sub-command
4	address, low 7 bits	Display address − 1
5	address, high bits	`00` for every address 1–128
6	colour code	`00` default; see below
7–38	32 characters	Two rows of 16, space-padded
39	`F7`	SysEx end

- **Address** — the same 1-based display address you set in the settings app, *minus one* on the wire. Hauptwerk performs this subtraction for you; a raw sender must do it itself. Address **128** is the settings channel (below), never a label.
- **Text** — exactly 32 characters: the first 16 are the top row, the next 16 the bottom row. Pad short text with spaces. By default the *Smart* label mode trims each row and treats a backslash \ as a line break; *Rigid* keeps both rows exactly (Section 3.5).
- **Colour** — `00` leaves the tab untinted; `01`–`07` select the Hauptwerk palette.

Code	Colour
00	default (no tint)
01	white
02	red
03	green
04	gold
05	blue
06	magenta
07	cyan

4.4.2 Example

To label the tab at display address **12** with *Vox Humana / 8'* in **red**, send these 39 bytes:

```
F0 7D 01 0B 00 02
56 6F 78 20 48 75 6D 61 6E 61 20 20 20 20 20
38 27 20 20 20 20 20 20 20 20 20 20 20 20 20
F7
```

The first line is the header (`0B` is address $12 - 1$; `02` is red), the next two lines are the two 16-character rows, and the last byte closes the frame.

4.4.3 Settings commands and reset

Two more frames are useful when driving the unit from your own software:

- **Settings command** — a normal 39-byte label frame sent to address **128**, with the command name in the top row and its value in the bottom row. This is the mechanism described in Section 4.3; the full command list is there.
- **Tabula restore** — the 4-byte frame `F0 7D 02 F7`. Send it once before your labels to return the unit to its built-in defaults (clear labels and colours, reset fonts, label modes, grid lines and input mode).

4.4.4 Sending the frames

Any tool that can transmit a raw SysEx file to the unit's MIDI output port (from your computer's point of view) will do. Free examples:

- **Windows** — MIDI-OX (its *Send SysEx* command).
- **macOS** — SysEx Librarian.
- **Linux** — `amidi -p` *`port`* `-s file.syx`.

Important: Always send the **Tabula restore** frame `F0 7D 02 F7` **first**, before any label or settings frames. The restore clears every label left over from the previous session; if it arrives late — or in the middle of a session — it wipes the labels you have just set and the tabs go blank. *Hauptwerk gets this ordering for free*, because it re-sends its displays whenever a sample set loads; a raw sender must send the restore itself, as the first frame of every run.

Note: Labels sent this way are **runtime-only**: they are not saved on the unit and are gone after it restarts, so they must be sent again each session.

4.5 Factory defaults

Setting	Default
Number of tabs	40
Number of panel widgets	5
Tab LCD32 addresses	1–40
Panel widget LCD32 addresses	41–45
Tab input binding	NRPN, channel 1, number = tab number −1
Tab output binding	NRPN, channel 2, number = tab number −1
Tab mode	ON/OFF
Physical input mode	Smart drag
LCD32 label display mode	Smart (tabs and widgets)
Tab / widget font	Noto Sans
Tab / widget font size	1.0
Kiosk grid lines	shown
Screen adjustment offsets	0 mm / 0 mm
Wi-Fi network	`Tabula-`*`XXXXXX`*
Wi-Fi password	on the registration card
Settings address	`192.168.4.1`

4.6 Troubleshooting

The settings page only shows a registration prompt. The unit is not registered. Contact support.

Nothing happens when I press a tab. Confirm the unit is registered, the correct output port is enabled in Hauptwerk, and the tab is not set to *Disabled*. Check the MIDI monitor while pressing a tab to see whether the unit is sending.

Hauptwerk does not receive my tab presses. Check the input binding on the tab (Learn it from Hauptwerk), and that Hauptwerk's MIDI input is enabled for the Tabula port.

The computer does not see a new MIDI device. Try another USB-A to USB-A cable and port. On Windows, if a device with the same USB identity was ever installed, uninstall it, then reconnect.

The tabs stay dark after a sample set loads. The unit was probably still starting when Hauptwerk loaded the sample set. Switch the unit on first, wait about 30 seconds for it to start, then load the sample set again.

The stop names are wrong or the text runs together. Try switching the LCD32 label display mode between *Smart* and *Rigid* (Section 3.5).

A tab needs a hard press, or triggers by itself. Recalibrate the buttons (Section 3.6).

Pressing one tab switches several. The physical input mode is set to *Smart drag* and you swept across several. Use *Toggle* if you prefer each tab fully independent.

I cannot reach the settings page. Join the `Tabula-`*`XXXXXX`* Wi-Fi network (use the password on the registration card) and browse to `192.168.4.1`.

The display does not line up with the tabs. Adjust it on the Screen Adjustment page (Section 3.9).